CLOUD COMPUTING

LAB EXAM

SUBMITTED TO:

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CLOUD COMPUTING

LAB EXAM

Q1 – AWS IAM Setup Using AWS CLI and Console Verification (10 marks)

You are working as a junior cloud engineer. Your team needs an IAM setup for the **Software Engineering** group. Using the **AWS CLI** and then verifying in the **AWS Management Console**, complete the following.

You must:

- Use **your own name** for the IAM user (for example: Ali, Sara, YOURNAME).
- Use **exactly** this group name: SoftwareEngineering.
- Take the specified CLI and console screenshots at each point.

1. **Create IAM group SoftwareEngineering using AWS CLI**
    - Use the AWS CLI to **create** an IAM group named SoftwareEngineering.

```
@23-22411-007-oss →/workspaces/Lab9 (main) $ aws iam create-group --group-name SoftwareEngineering
{
    "Group": {
        "Path": "/",
        "GroupName": "SoftwareEngineering",
        "GroupId": "AGPATFX3NIVRCC3HA4SDE",
        "Arn": "arn:aws:iam::218496845154:group/SoftwareEngineering",
        "CreateDate": "2026-01-19T07:37:05+00:00"
    }
}
@23-22411-007-oss →/workspaces/Lab9 (main) $
```

    - **Save screenshot as:** q1_create_group.png — terminal showing the **create-group command** and its **output**.
    - Then use the AWS CLI to **view the group details**.

```
@23-22411-007-oss →/workspaces/Lab9 (main) $ aws iam get-group --group-name SoftwareEngineering
{
    "Users": [],
    "Group": {
        "Path": "/",
        "GroupName": "SoftwareEngineering",
        "GroupId": "AGPATFX3NIVRCC3HA4SDE",
        "Arn": "arn:aws:iam::218496845154:group/SoftwareEngineering",
        "CreateDate": "2026-01-19T07:37:05+00:00"
    }
}
@23-22411-007-oss →/workspaces/Lab9 (main) $
```

    - **Save screenshot as:** q1_group_details.png — terminal showing output that includes:
        - The **group name** SoftwareEngineering, and
        - The **group ARN**.

2. **Create IAM user (your name) and view details**
    - Using the AWS CLI, **create an IAM user** whose username is based on your own name (for example, Ali, Sara, YOURNAME).

```
@23-22411-007-oss ➜/workspaces/Lab9 (main) $ aws iam create-user --user-name Amina
{
    "User": {
        "Path": "/",
        "UserName": "Amina",
        "UserId": "AIDATFX3NIVRAP2224JI7",
        "Arn": "arn:aws:iam::218496845154:user/Amina",
        "CreateDate": "2026-01-19T07:39:52+00:00"
    }
}
@23-22411-007-oss ➜/workspaces/Lab9 (main) $
```

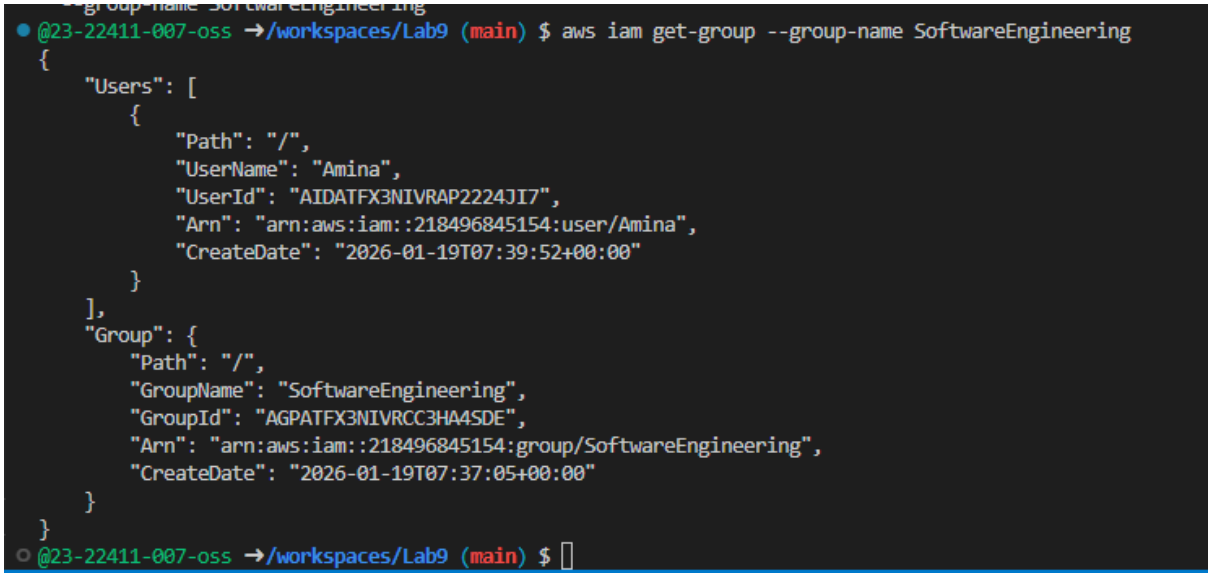
    - **Save screenshot as:** q1_create_user.png — terminal showing the **create-user command** and its **output**.
    - Use the AWS CLI to **view the user details**.

```
@23-22411-007-oss ➜/workspaces/Lab9 (main) $ aws iam get-user --user-name Amina
{
    "User": {
        "Path": "/",
        "UserName": "Amina",
        "UserId": "AIDATFX3NIVRAP2224JI7",
        "Arn": "arn:aws:iam::218496845154:user/Amina",
        "CreateDate": "2026-01-19T07:39:52+00:00"
    }
}
@23-22411-007-oss ➜/workspaces/Lab9 (main) $
```

    - **Save screenshot as:** q1_user_details.png — terminal showing output that includes:
        - The **username**, and
        - The **user ARN**.
3. **Add the IAM user to the SoftwareEngineering group**
    - Using the AWS CLI, **add the IAM user** (created above) to the SoftwareEngineering group.

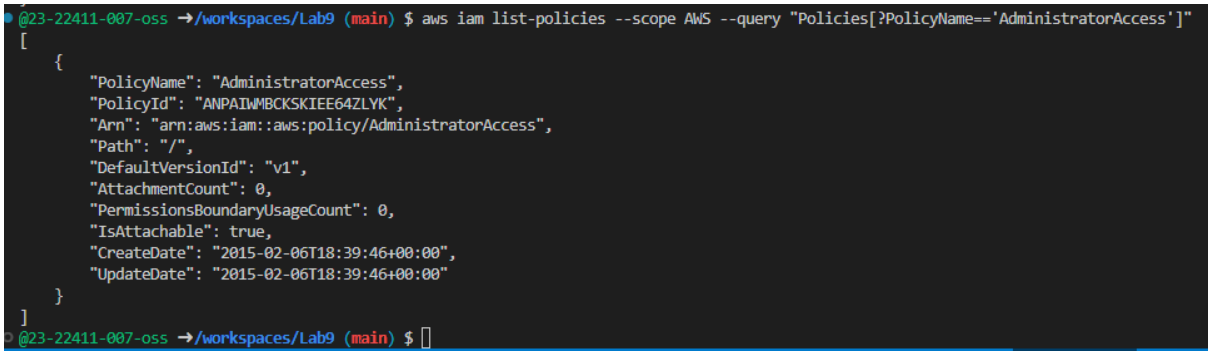
```
@23-22411-007-oss ➜/workspaces/Lab9 (main) $ aws iam add-user-to-group \
  --user-name Amina \
  --group-name SoftwareEngineering
```

    - **Save screenshot as:** q1_add_user_to_group.png — terminal showing the **add-user-to-group command** and its **output** (or lack of error).
    - Use the AWS CLI to **view the group membership** and confirm your user is a member.

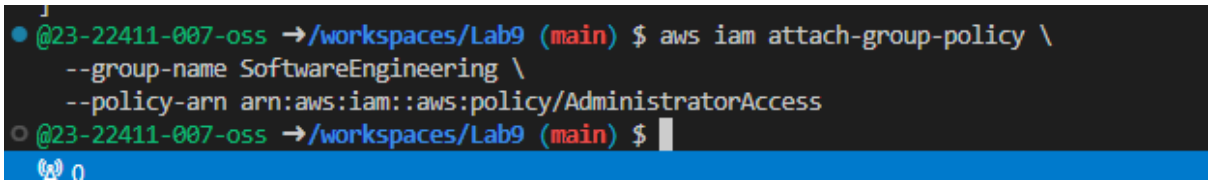


- **Save screenshot as:** q1_group_membership.png — terminal showing your IAM user listed as a **member** of SoftwareEngineering.

4. **Attach AdministratorAccess managed policy to the SoftwareEngineering group**
   - Using the AWS CLI, **locate and identify** the AWS managed policy named AdministratorAccess, including its ARN (for example, by using aws iam list-policies or aws iam get-policy with the known ARN).



   - **Save screenshot as:** q1_find_admin_policy.png — terminal showing the command(s) and output that clearly include:
     - The **policy name** AdministratorAccess, and
     - The **policy ARN**.
   - Using the AWS CLI, **attach** the AdministratorAccess managed policy to the SoftwareEngineering group using its policy ARN.



- **Save screenshot as:** q1_attach_admin_policy.png — terminal showing the **attach-group-policy command** and confirmation (no error).

5. **List attached policies of the SoftwareEngineering group**

   - Using the AWS CLI, **list the policies** attached to the SoftwareEngineering group (for example, list-attached-group-policies).

```
@23-22411-007-oss →/workspaces/Lab9 (main) $ aws iam list-attached-group-policies \
  --group-name SoftwareEngineering
{
    "AttachedPolicies": [
        {
            "PolicyName": "AdministratorAccess",
            "PolicyArn": "arn:aws:iam::aws:policy/AdministratorAccess"
        }
    ]
}
@23-22411-007-oss →/workspaces/Lab9 (main) $
```

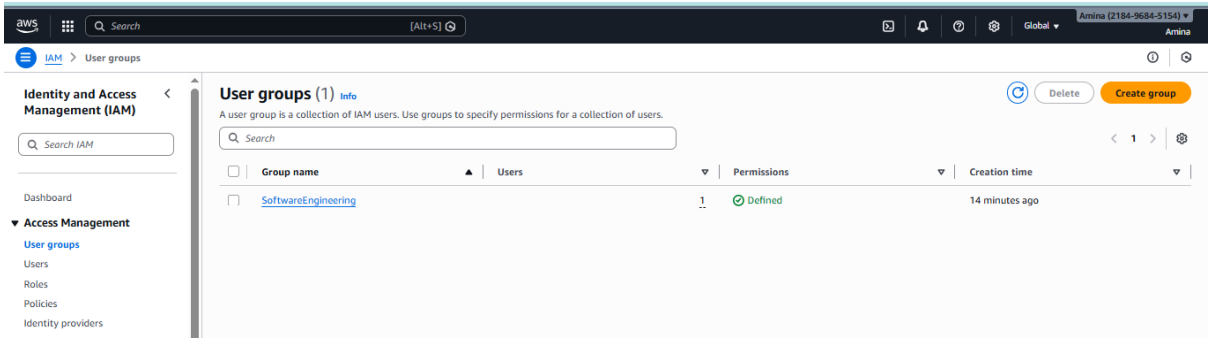
   - **Save screenshot as:** q1_list_group_policies.png — terminal showing the output that clearly includes AdministratorAccess attached to SoftwareEngineering.

6. **Verify IAM configuration in AWS Management Console**

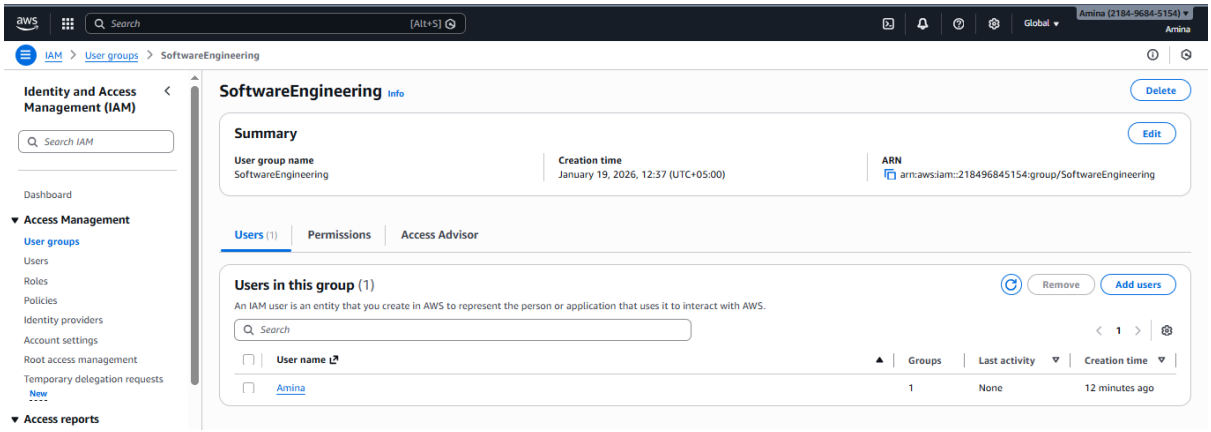
After completing all CLI steps above, log in to the **AWS Management Console** and verify that:

- The SoftwareEngineering group exists.
- Your IAM user exists and is part of that group.
- The group has the AdministratorAccess managed policy attached.

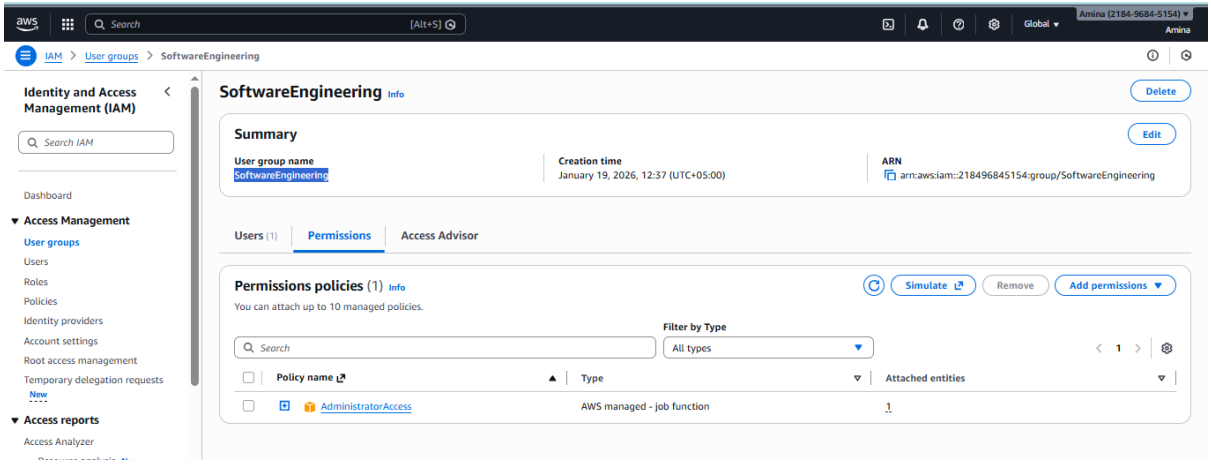
Take the following screenshots:



   - **Save screenshot as:** q1_console_group.png — IAM console page clearly showing the **SoftwareEngineering group**.



- **Save screenshot as:** q1_console_user_in_group.png — IAM console page clearly showing your **IAM user** and that it belongs to the SoftwareEngineering group.



- **Save screenshot as:** q1_console_group_policy.png — IAM console page for the SoftwareEngineering group clearly showing the **AdministratorAccess** managed policy is attached.

Q2 – Terraform Lab: Simple AWS Environment with Nginx over HTTPS (30 marks)

You are working as an infrastructure engineer and must provision a simple application environment in AWS using **Terraform only**. No resources may be created manually in the AWS console.

You will create Terraform configuration and shell script files to deploy the required infrastructure.

You must:

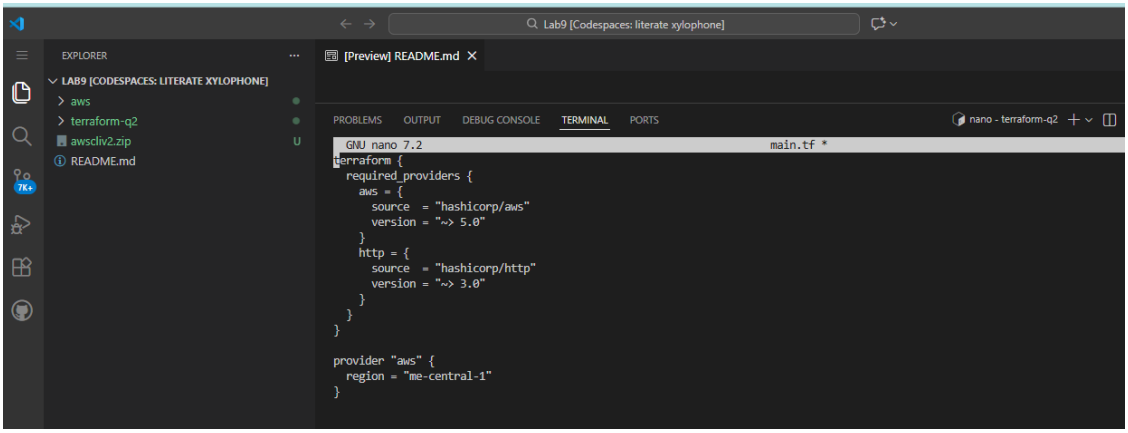
- Write all Terraform files and scripts yourself.
- Use a data "http" source and a locals block for your /32 IP.

1. **Configure the AWS provider**

Create the provider configuration so that the AWS provider reads from:

- ~/.aws/config
- ~/.aws/credentials

(You may use the default profile or a named profile.)



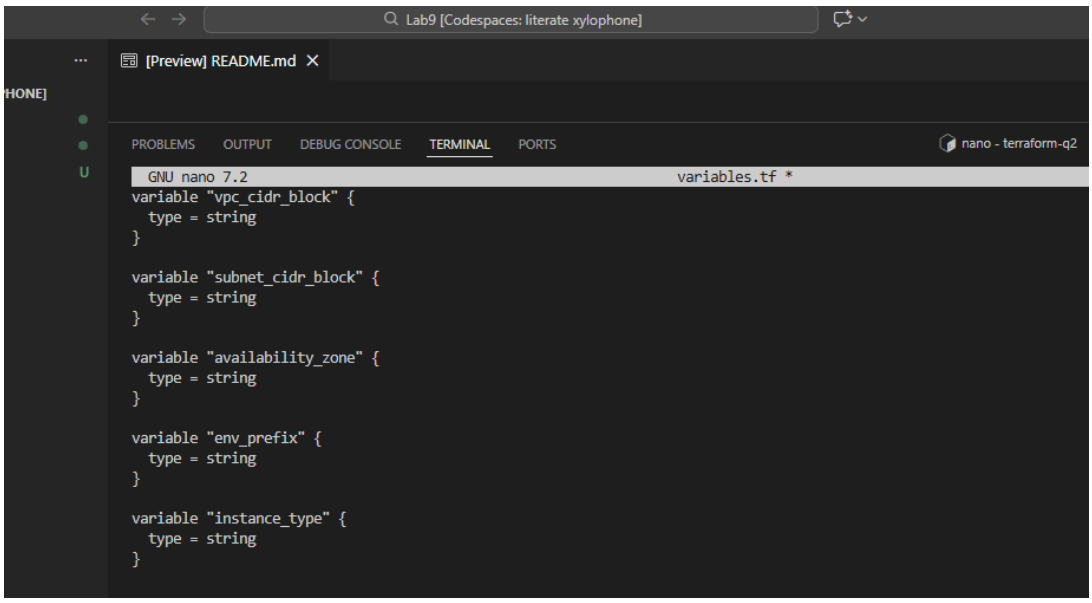
- **Save screenshot as:** q2_provider.png — editor or terminal view clearly showing the **provider** block in your .tf file (e.g., main.tf).

2. **Define input variables**

Define the following input variables:

- vpc_cidr_block
- subnet_cidr_block
- availability_zone
- env_prefix
- instance_type

Place them in variables.tf or any .tf file.



- **Save screenshot as:** q2_variables.png — editor view showing all five variable definitions.

3. **Create VPC and subnet**

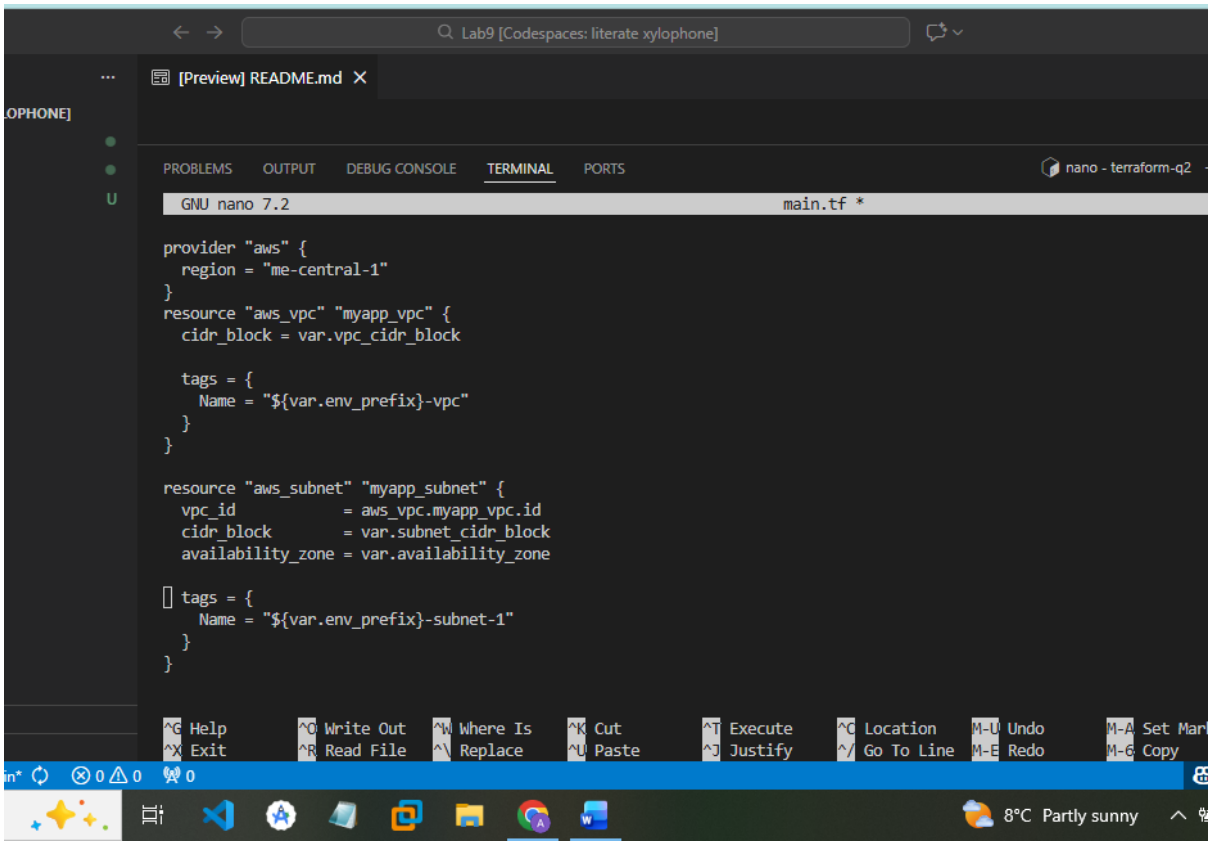
Using Terraform:

- Create a **VPC** (e.g., aws_vpc.myapp_vpc) using var.vpc_cidr_block as CIDR. Tag:

Name = "<env_prefix>-vpc"

- Create a **subnet** attached to this VPC with:
  - var.subnet_cidr_block as CIDR,
  - var.availability_zone as AZ,
  - Tag:

Name = "<env_prefix>-subnet-1"



- **Save screenshot as:** q2_vpc_subnet.png — editor view showing VPC and subnet resources with tags.

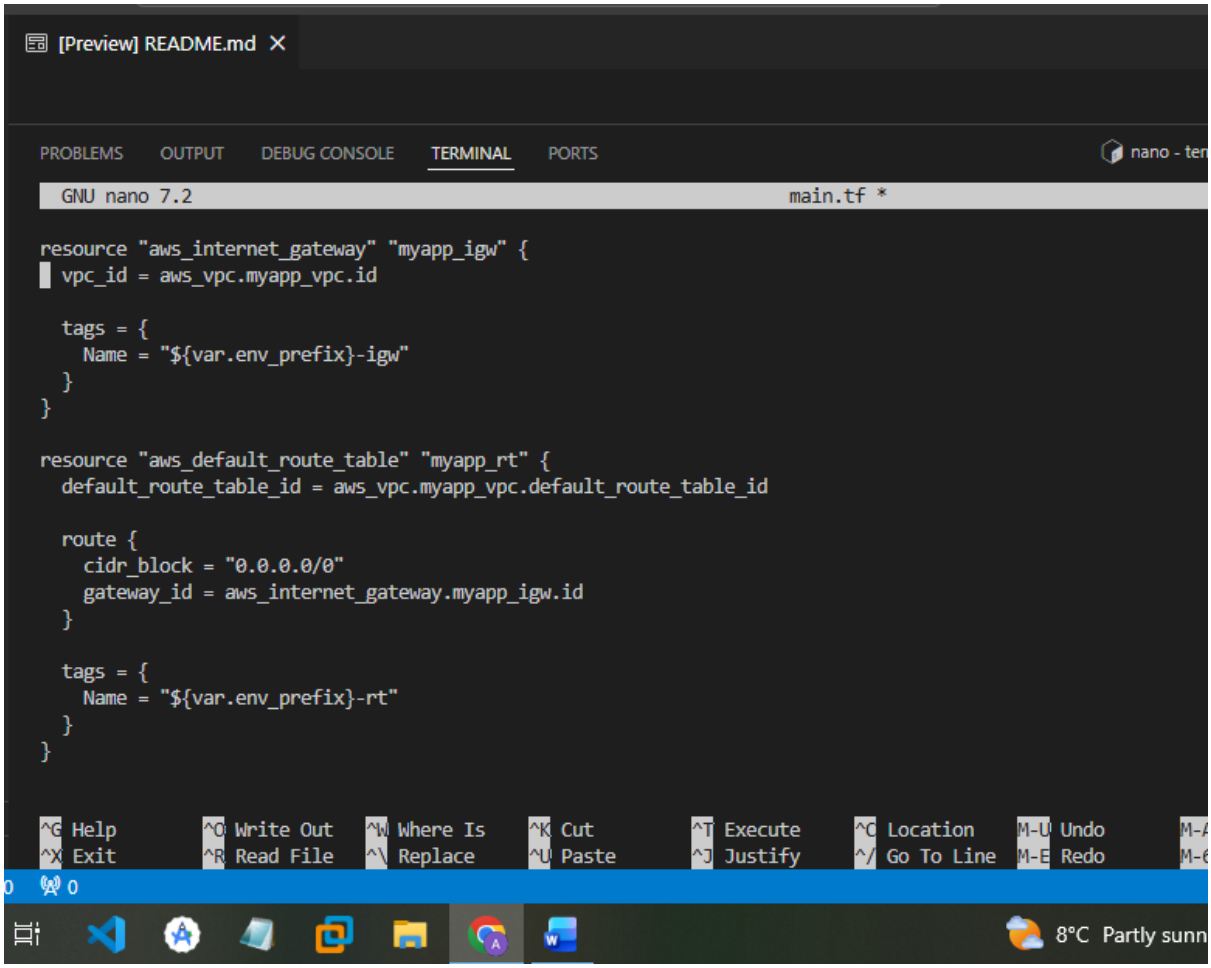
4. **Create Internet Gateway and configure default route table**

- Create an **Internet Gateway** attached to the VPC and tag it:

Name = "<env_prefix>-igw"

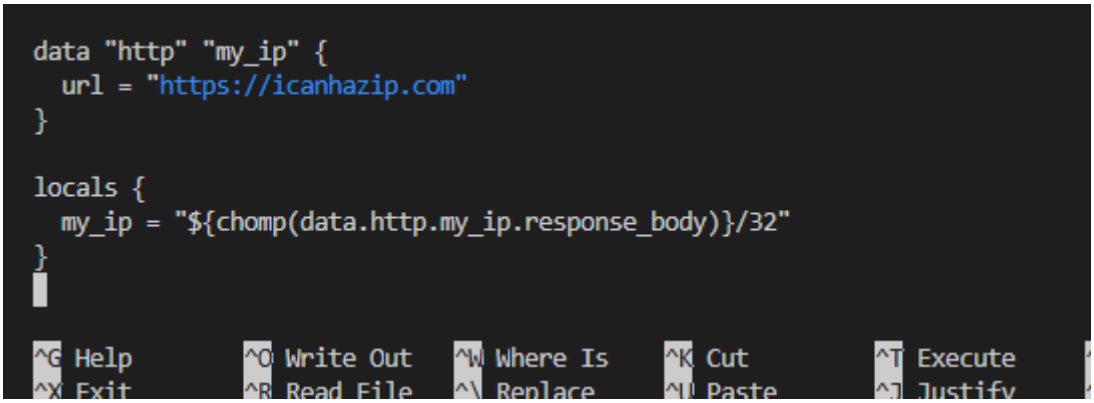
- Use the VPC's **default route table** to add a route 0.0.0.0/0 pointing to this IGW, and tag:

Name = "<env_prefix>-rt"



- **Save screenshot as:** q2_igw_route_table.png — editor view showing the IGW and default route table configuration, including the 0.0.0.0/0 route and tags.

5. **Discover public IP and compute /32 CIDR using data + locals**
   - Use a Terraform data "http" block to query your **current public IP** (e.g., https://icanhazip.com).
   - Use a locals block to:
     - Take the raw HTTP response body,
     - Strip trailing newline(s),
     - Append /32 to create a single-host CIDR locals.my_ip.



- **Save screenshot as:** q2_http_and_locals.png — editor view showing the data "http" block and locals block with locals.my_ip.

6. **Configure the default security group in the VPC**

Manage the **default security group** of your VPC with:

- Ingress:
  - SSH (TCP 22) from locals.my_ip.
  - HTTP (TCP 80) from 0.0.0.0/0.
  - HTTPS (TCP 443) from 0.0.0.0/0.
- Egress:
  - All outbound traffic to 0.0.0.0/0.

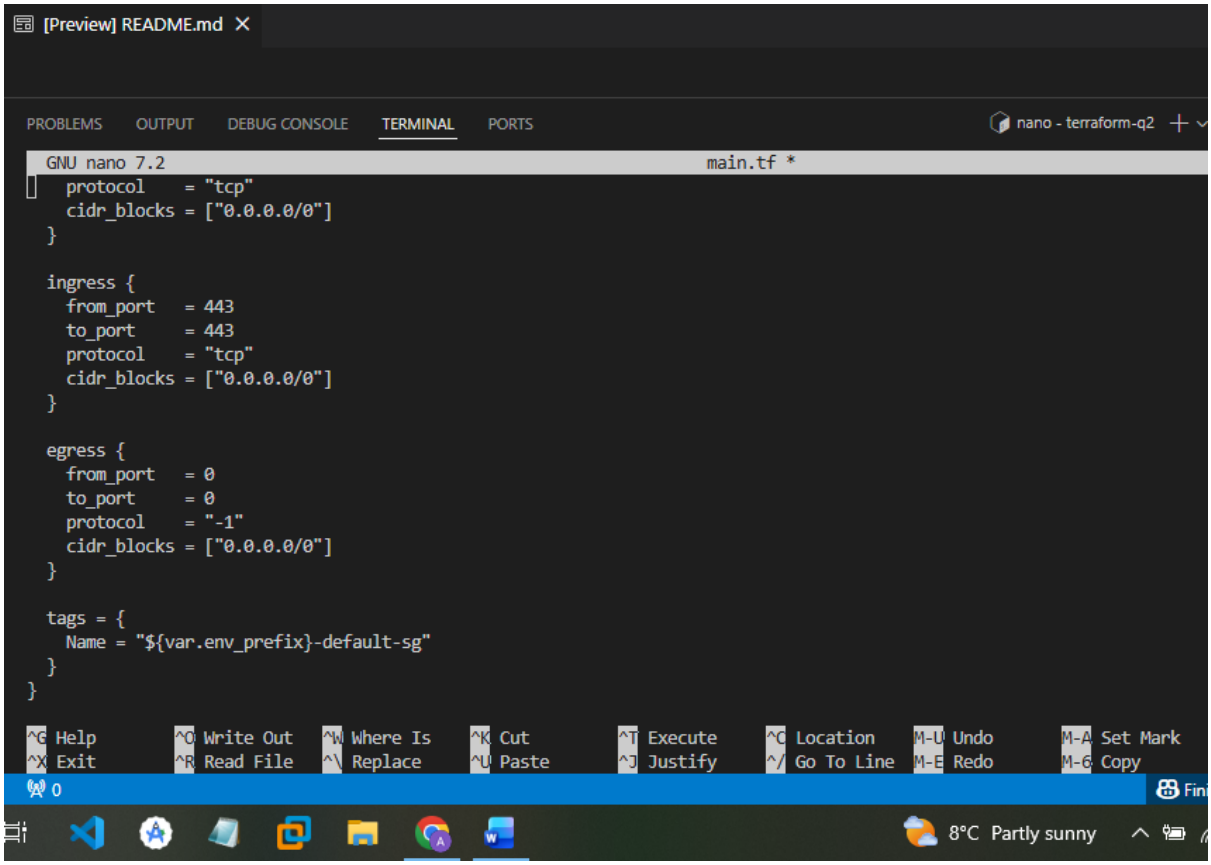
Tag:

Name = "<env_prefix>-default-sg"

Code (security Group):

```
resource "aws_default_security_group" "default_sg" {
 vpc_id = aws_vpc.myapp_vpc.id
 ingress {
  from_port   = 22
  to_port     = 22
  protocol    = "tcp"
  cidr_blocks = [locals.my_ip]
 }
 ingress {
  from_port   = 80
  to_port     = 80
```

```
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  ingress {
    from_port   = 443
    to_port     = 443
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
  tags = {
    Name = "${var.env_prefix}-default-sg"
  }
}
```



- **Save screenshot as:** q2_default_sg.png — editor view showing the default SG Terraform resource with all rules and tag.

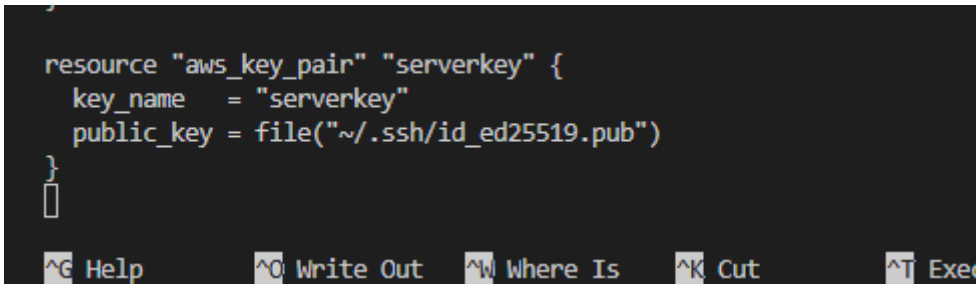
7. **Create an AWS key pair for SSH**

Create an aws_key_pair resource:

- Key name: serverkey
- Public key: ~/.ssh/id_ed25519.pub

CODE:

```
resource "aws_key_pair" "serverkey" {
  key_name   = "serverkey"
  public_key = file("~/.ssh/id_ed25519.pub")
}
```



- **Save screenshot as:** q2_keypair.png — editor view showing the key pair resource referencing ~/.ssh/id_ed25519.pub.

8. **Create the EC2 instance resource**

Launch an EC2 instance that:

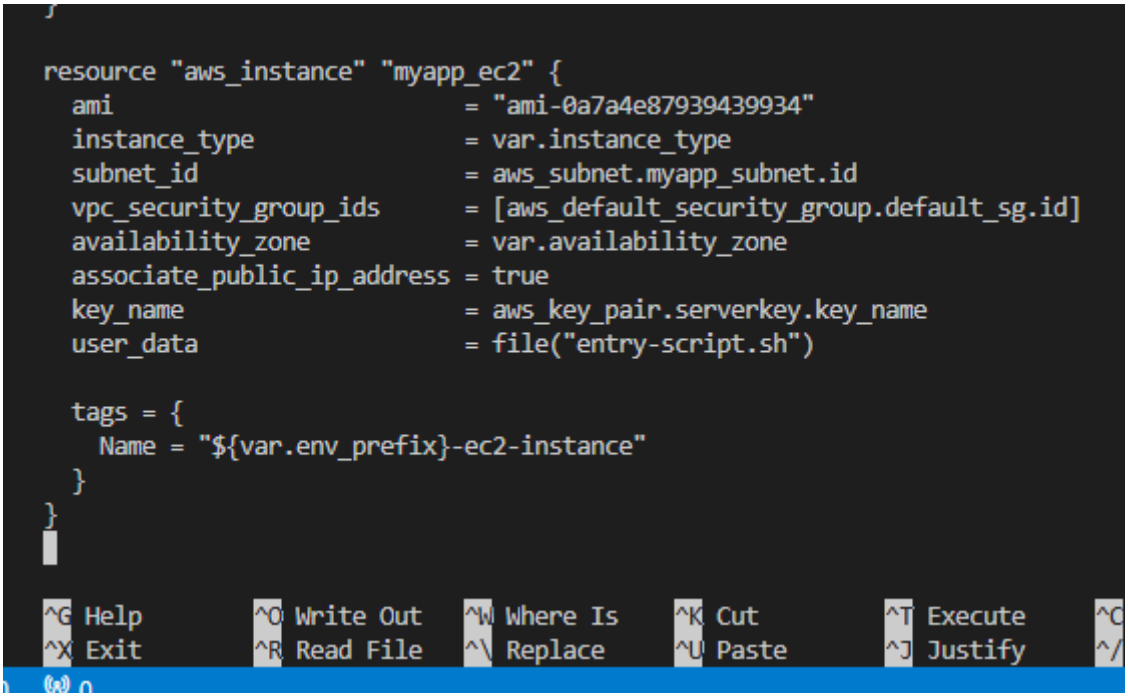
- Uses an Amazon Linux 2023 AMI (hard-coded ID).
- Uses var.instance_type.
- Is in the subnet you created.
- Attaches the **default security group** from step 6.
- Uses var.availability_zone.
- Has a public IP address.
- Uses key pair serverkey.
- Uses user_data from a local file entry-script.sh.
- Is tagged:

Name = "<env_prefix>-ec2-instance"

CODE:

```
resource "aws_instance" "myapp_ec2" {
  ami                    = "ami-0a7a4e87939439934"
  instance_type          = var.instance_type
  subnet_id              = aws_subnet.myapp_subnet.id
  vpc_security_group_ids = [aws_default_security_group.default_sg.id]
  availability_zone      = var.availability_zone
  associate_public_ip_address = true
  key_name               = aws_key_pair.serverkey.key_name
  user_data              = file("entry-script.sh")

  tags = {
    Name = "${var.env_prefix}-ec2-instance"
  }
}
```



- **Save screenshot as:** q2_ec2_resource.png — editor view showing the EC2 instance resource with subnet, SG, key pair, user_data, and tag.

9. **Create entry-script.sh to configure Nginx + HTTPS**

In the same directory as your .tf files, create entry-script.sh that:

- Installs **Nginx**.
- Generates/configures a **self-signed TLS certificate**.
- Configures Nginx to:
  - Listen on **HTTPS (443)** using that certificate,
  - Serve **HTTP (80)** (redirect to HTTPS or serve a page).
- Enables Nginx on boot and ensures it is running after the script.
- Serves a page that includes:
  - Your **name**, and
  - The word **"Terraform"** (e.g. This is YOURNAME's Terraform environment.).

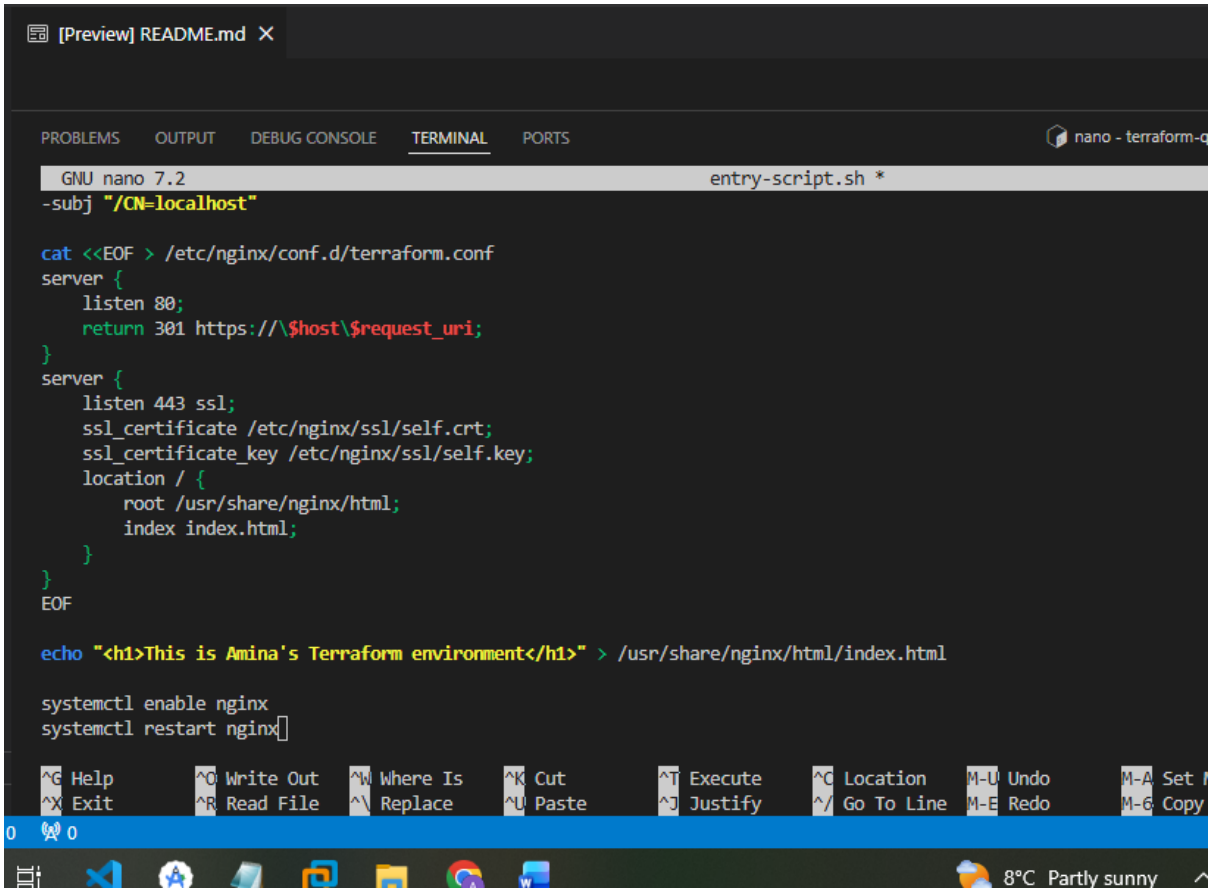
CODE:

```
#!/bin/bash

dnf install -y nginx openssl

mkdir -p /etc/nginx/ssl
```

```
openssl req -x509 -nodes -days 365 \
-newkey rsa:2048 \
-keyout /etc/nginx/ssl/self.key \
-out /etc/nginx/ssl/self.crt \
-subj "/CN=localhost"
cat <<EOF > /etc/nginx/conf.d/terraform.conf
server {
   listen 80;
   return 301 https://\$host\$request_uri;
}
server {
   listen 443 ssl;
   ssl_certificate /etc/nginx/ssl/self.crt;
   ssl_certificate_key /etc/nginx/ssl/self.key;
   location / {
      root /usr/share/nginx/html;
      index index.html;
   }
}
EOF
echo "<h1>This is Amina's Terraform environment</h1>" > /usr/share/nginx/html/index.html
systemctl enable nginx
systemctl restart nginx
```



- **Save screenshot as:** q2_entry_script.png — editor or terminal view showing the full content of entry-script.sh.

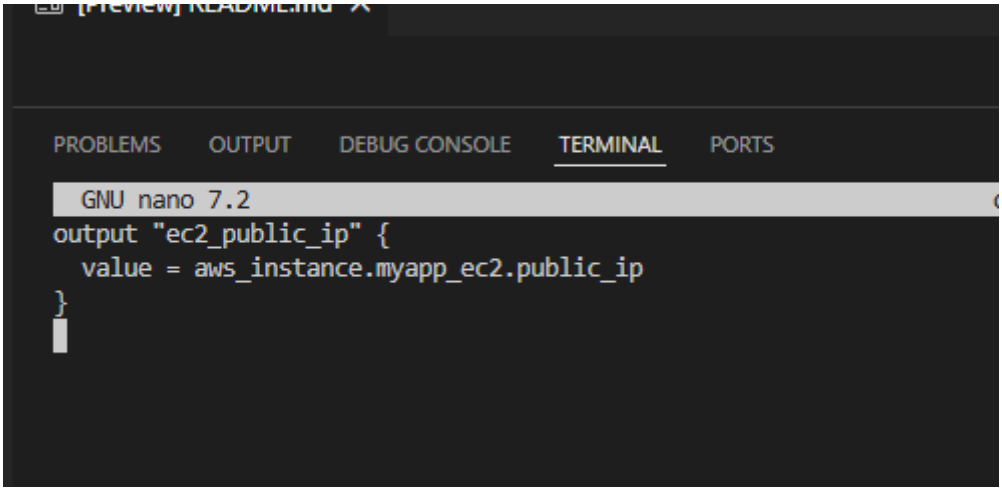
10. **Add Terraform output for public IP**

Add an output (e.g. ec2_public_ip) to print the instance's public IP after terraform apply.

output "ec2_public_ip" {

value = aws_instance.myapp_ec2.public_ip

}



- **Save screenshot as:** q2_output_block.png — editor view showing the Terraform output exposing the EC2 public IP.

11. **Set variable values for apply time**

When running terraform apply, use:

vpc_cidr_block = "10.0.0.0/16"

subnet_cidr_block = "10.0.10.0/24"

availability_zone = "me-central-1a"

env_prefix = "dev"

instance_type = "t3.micro"

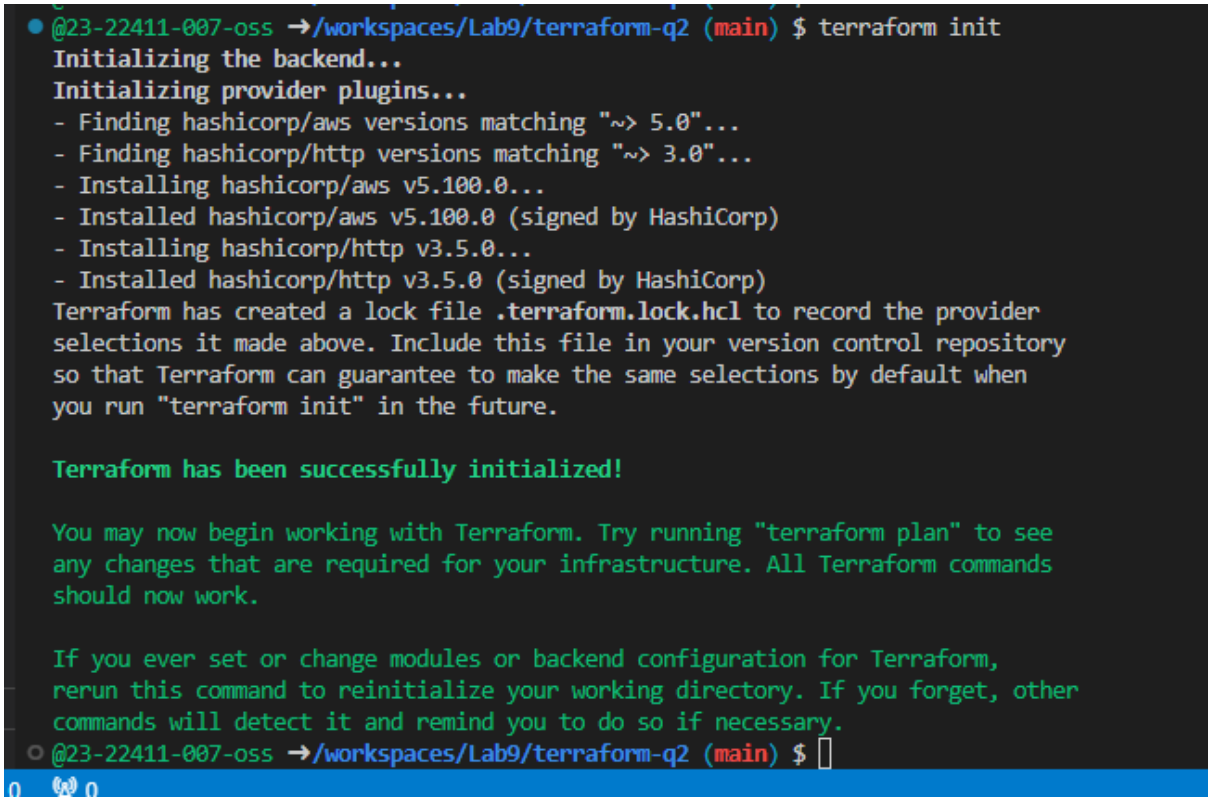
(via terraform.tfvars, -var flags, or defaults)

- **Save screenshot as:** q2_tfvars_or_vars.png — editor view of terraform.tfvars or CLI evidence that these values are used.

12. **Run Terraform commands and capture outputs**

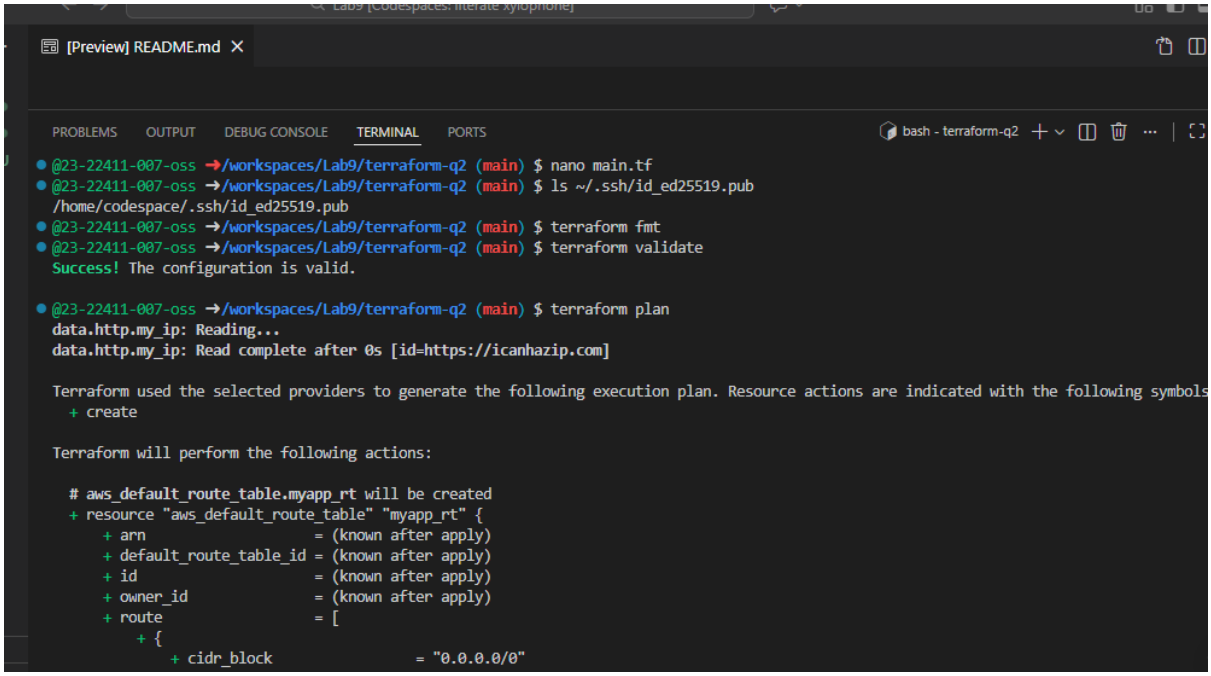
From the Terraform project directory:

terraform init



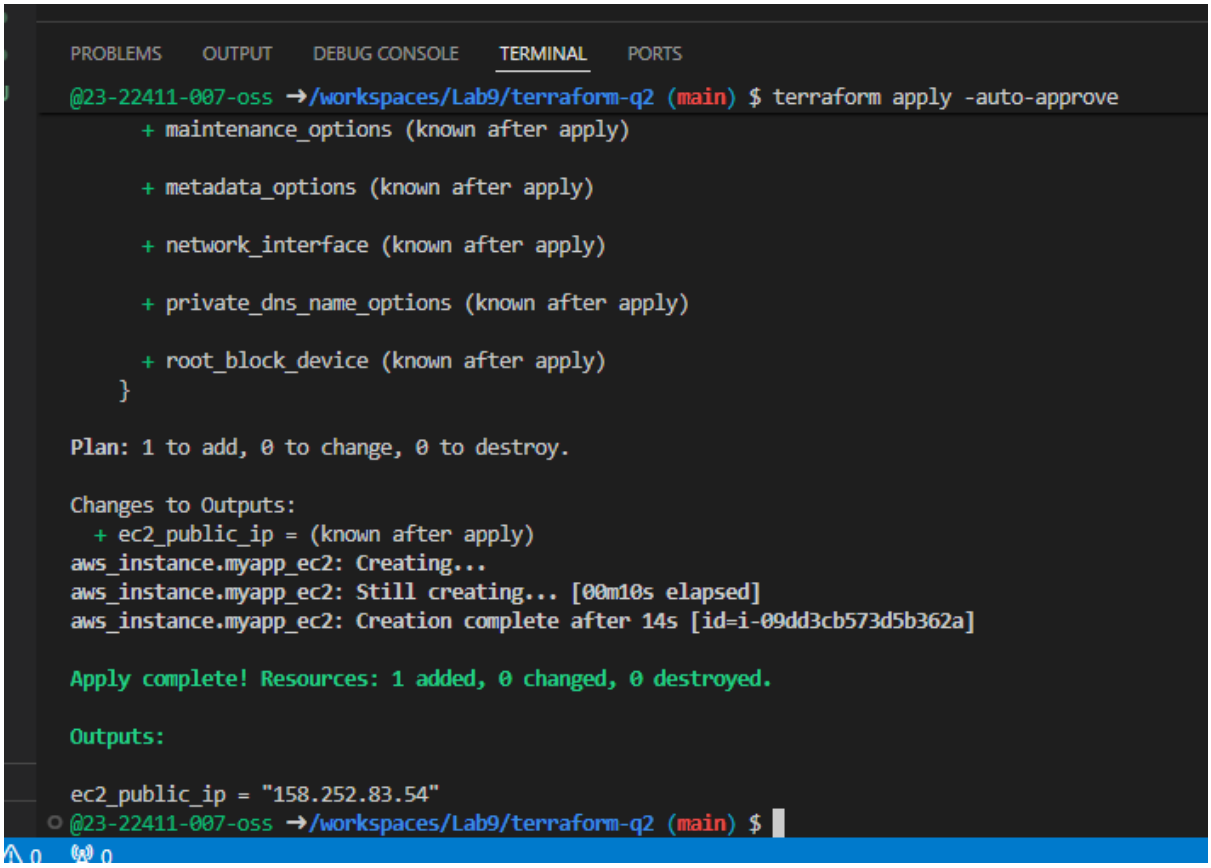
- **Save screenshot as:** q2_terraform_init.png — terminal showing terraform init and successful initialization.

terraform plan



- **Save screenshot as:** q2_terraform_plan.png — terminal showing terraform plan and part of the plan.

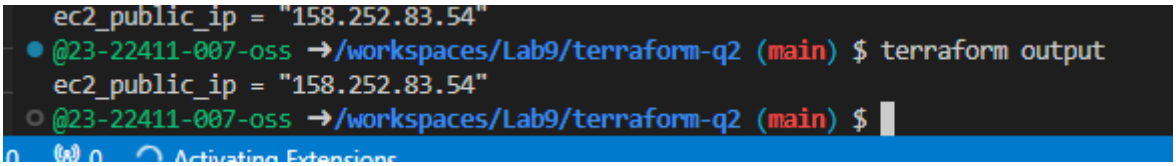
terraform apply



(You may use -auto-approve.)

- **Save screenshot as:** q2_terraform_apply.png — terminal showing terraform apply and successful resource creation.

terraform output

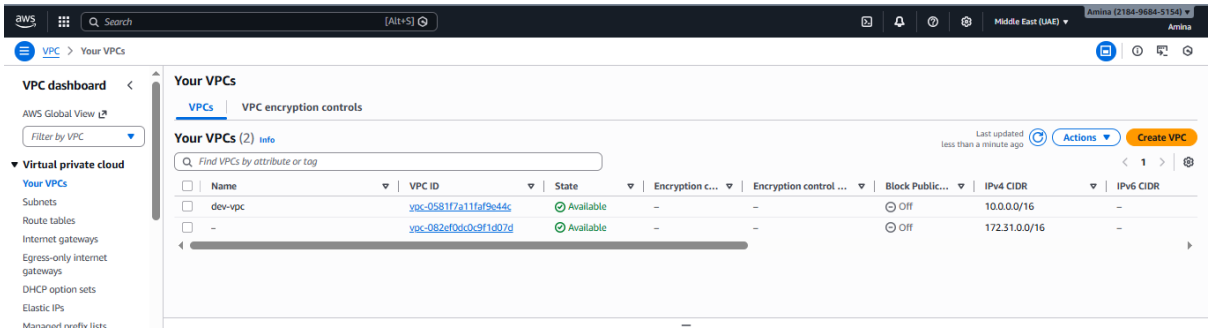


- **Save screenshot as:** q2_terraform_output.png — terminal showing outputs including the EC2 public IP.

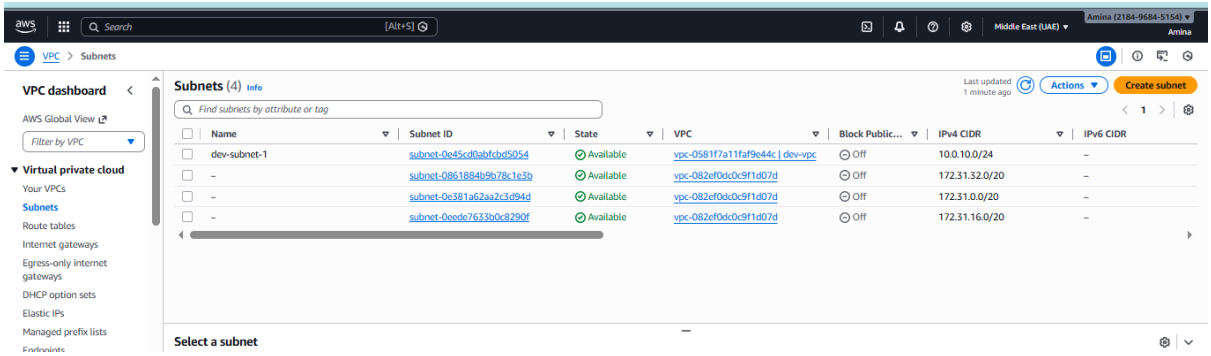
13. **Verify Terraform resources in AWS console**

Use the AWS console to confirm:

- **VPC and Subnet**

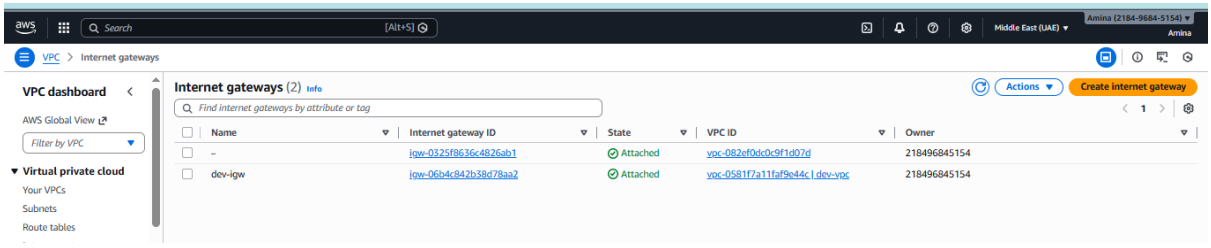


- **Save screenshot as:** q2_console_vpc.png — VPC with CIDR 10.0.0.0/16 and tag Name = dev-vpc.

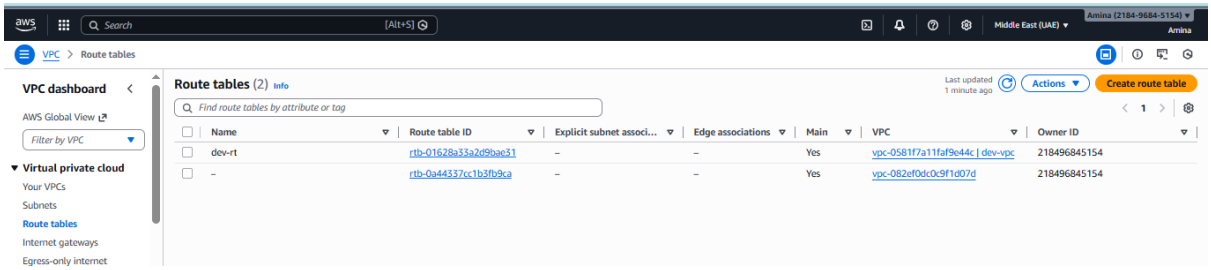


- **Save screenshot as:** q2_console_subnet.png — subnet with CIDR 10.0.10.0/24, AZ me-central-1a, tag Name = dev-subnet-1.

- **Internet Gateway and Route Table**

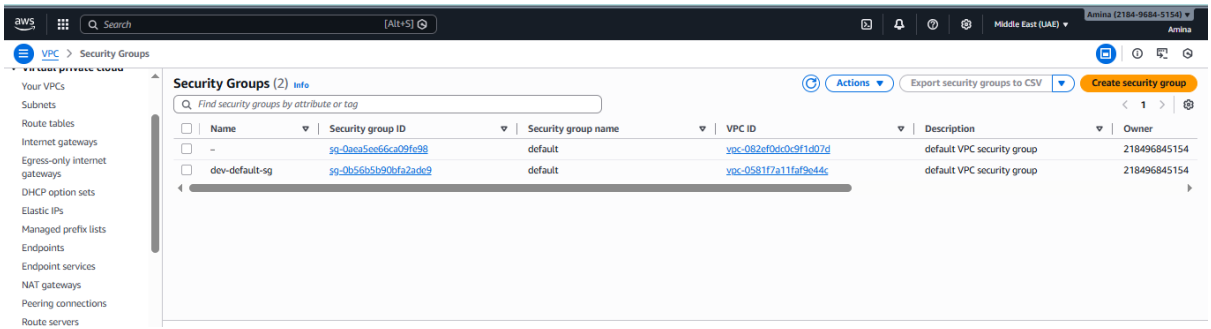


- **Save screenshot as:** q2_console_igw.png — IGW attached to the VPC with tag Name = dev-igw.



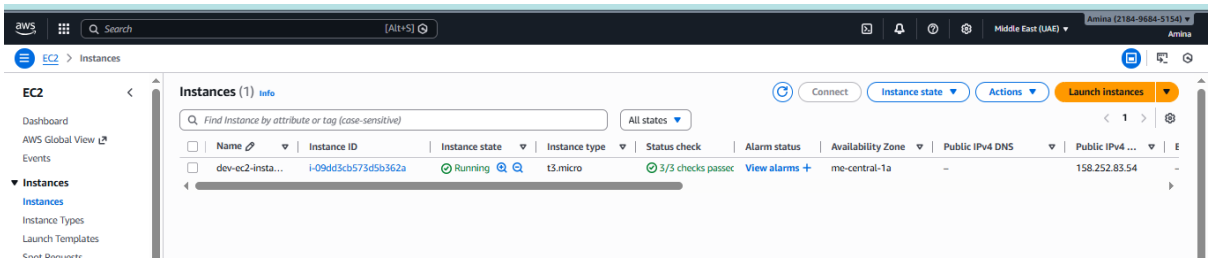
- **Save screenshot as:** q2_console_route_table.png — route table with tag Name = dev-rt and route 0.0.0.0/0 to the IGW.

- **Security Group**



- **Save screenshot as:** q2_console_sg.png — default SG showing:
  - SSH (22) from your /32 host,
  - HTTP (80) from 0.0.0.0/0,
  - HTTPS (443) from 0.0.0.0/0,
  - Egress all to 0.0.0.0/0,
  - Tag Name = dev-default-sg.

- **EC2 instance**



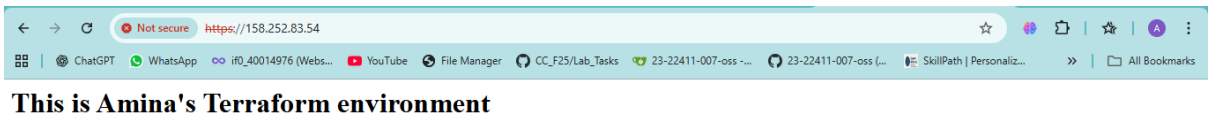
- **Save screenshot as:** q2_console_ec2.png — instance:
  - In subnet 10.0.10.0/24, AZ me-central-1a,
  - With a public IP,
  - Using key pair serverkey,
  - Tagged Name = dev-ec2-instance.

14. **Verify HTTPS access from browser**

From your local machine:

https://<public-ip-of-instance>

Accept any self-signed cert warning and verify the page content.



- **Save screenshot as:** q2_https_browser.png — browser screenshot showing:
  - Address bar https://<public-ip-of-instance>,
  - Nginx page with your **name** and **"Terraform"**.

Q3 – Ansible Playbook for EC2 Web Server Using Q2 Instance (10 marks)

You are working as a DevOps engineer. A previous solution used a shell script on an Amazon Linux EC2 instance to:

- Update the system,
- Install and start Apache HTTPD,
- Fetch instance metadata (public IP and public hostname) using **IMDSv2**,
- Print the public IP,
- Restart the HTTPD service.

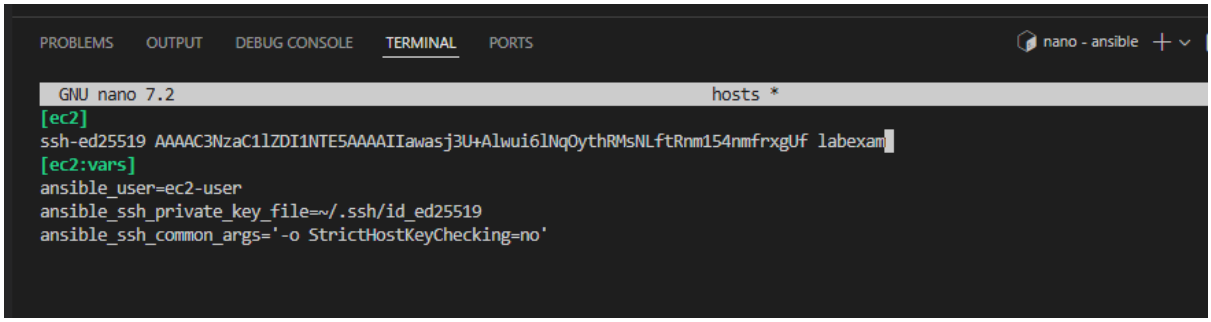
Your task is to convert this logic into an **Ansible-based** solution targeting the EC2 instance created in **Step 2 (Q2)**.

If Nginx is running from Q2's user data, you must **stop/disable or uninstall nginx** so that httpd can bind to port 80.

1. **Create Ansible inventory file hosts**

In a directory such as Lab_exam/ansible/, create hosts that:

- Defines group ec2 with your Q2 instance's **public IP**.
- Sets group vars:
  - ansible_user = ec2-user
  - ansible_ssh_private_key_file = ~/.ssh/id_ed25519
  - ansible_ssh_common_args to disable strict host key checking (-o StrictHostKeyChecking=no).



- **Save screenshot as:** q3_hosts.png — editor or terminal view showing the complete hosts file (group, IP, vars).

2. **Create project-level ansible.cfg**

In the same directory, create ansible.cfg that:

- Disables host key checking.
- Sets remote Python interpreter to /usr/bin/python3.
- (Optional) Sets inventory = ./hosts.

- **Save screenshot as:** q3_ansible_cfg.png — editor view showing ansible.cfg with required settings.

3. **Create Ansible playbook (e.g. my-playbook.yml)**

Create my-playbook.yml that:

- Targets group ec2.
- Uses become: true.
- Performs actions equivalent to the shell script:
  - Update system packages.
  - Stop/disable or uninstall nginx if present (to free port 80).
  - Install httpd.
  - Start and enable httpd service.
  - Get IMDSv2 token using uri:
    - URL: http://169.254.169.254/latest/api/token
    - Method: PUT
    - Header: X-aws-ec2-metadata-token-ttl-seconds: 21600
  - Use token to fetch:
    - http://169.254.169.254/latest/meta-data/public-ipv4
    - http://169.254.169.254/latest/meta-data/public-hostname
  - Print **public IP** with debug.
  - Restart httpd.

- **Save screenshot as:** q3_playbook.png — editor view showing the full contents of my-playbook.yml.

4. **Run the Ansible playbook**

Execute:

ansible-playbook -i hosts my-playbook.yml

(or without -i if your ansible.cfg sets the inventory.)

- **Save screenshot as:** q3_play_run.png — terminal showing:
  - The ansible-playbook command,
  - All tasks (update, nginx handling, httpd, IMDSv2, debug of public IP),
  - Successful completion.

5. **Verify HTTP access**

From your local machine:

http://<public-ip-ec2>

Confirm Apache (httpd) is responding.

- **Save screenshot as (optional):** q3_http_browser.png — browser or curl output showing HTTP access to http://<public-ip-ec2>.

Files Required (Q3) in Lab_exam repo, e.g. Lab_exam/ansible/:

- hosts
- ansible.cfg
- my-playbook.yml

Cleanup (ungraded)

After collecting all evidence:

1. From your Terraform project directory (Q2):

terraform destroy -auto-approve

- **Save screenshot as (recommended):** cleanup_terraform_destroy.png — terminal showing destroy output.

2. In AWS console, verify that no lab-related EC2 instances remain running.

- **Save screenshot as (recommended):** cleanup_ec2_console.png — EC2 console showing no remaining lab instances.

(If your instructor gives different cleanup instructions, follow those.)